



Mrityunjay Shukla
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Examination	University	Institute	Year	CPI/%
Ph.D. (CSE dept)	IIT Bombay	IIT Bombay	2024-	8.83
Graduation (ECE dept)	MIT, Manipal	Manipal, Karnataka	2020-24	8.23
Intermediate/+2	CBSE	DPS, Kalyanpur	2019-20	87
Matriculation	CBSE	DPS, Kalyanpur	2018	93

PUBLICATIONS

- **M.Shukla***, S. Roy, S. Saha, B. Panda. **"PRowhammer: Propagating bit-flips from CPU to GPU"** Accepted at the 53rd IEEE/ACM Annual International Symposium on Computer Architecture International Symposium on Computer Architecture (ISCA '26).
- S. Ahamed*, **M.Shukla**, K. Singla, S. Saha. **"Everything Depends on Your Hammer: A Systematic Rowhammer Attack Exploration on SPHINCS+"** In Proceedings of the 44th IEEE/ACM International Conference On Computer Aided Design (ICCAD '25)

* denotes primary author.

RESEARCH WORK

- **Designing GPU microarchitecture for ray tracing**
(Ongoing Research | Guide: **Prof. Biswabandan Panda**) (Nov'25-Present)
 - Developed a comprehensive understanding of GPU-based ray tracing by analyzing state-of-the-art scheduling and prefetching mechanisms.
 - Quantitatively evaluated the interplay between scheduling and prefetching techniques to determine whether their combined effects enhance or degrade overall performance.

COURSE WORK

Course Code	Course Name	Grade
CS 683	Advanced Computer Architecture	AA
CS 773	Computer Architecture for Performance and Security	AA
CS 6102	Implementation Security in Cryptography	AB
CS 725	Foundations of Machine Learning	BB
CS 744	Design and Engineering of Computing Systems	BC
CS 219	Operating Systems	BC
CS 490	R & D Project	AA

COURSE PROJECTS

- **Rowhammer attack on discrete GPUs**
(CS 490: R&D Project | **Prof. Biswabandan Panda** and **Prof. Sayandeep Saha**) (Aug'24-Dec'24)
 - Studied how **Rowhammer** attacks are mounted on **CPU DRAM**, including reverse-engineering physical-to-DRAM address mappings to build the foundational understanding needed to investigate similar attacks on **GPU VRAMs**.
 - Analyzed **NVIDIA GPU architecture (RTX 4090 and RTX 3050)** and attempted to reverse-engineer their physical-to-DRAM address mappings using the **PTX discard instruction** as a step toward assessing **Rowhammer** vulnerabilities on GPUs.
- **Performance Optimization**
(CS 683: Advanced Computer Architecture | **Prof. Biswabandan Panda**) (Aug'24-Dec'24)
 - Optimized 2D-convolution operation. Achieved 4-5x speedup using **blocked matrix multiplication, SIMD & software prefetching**.
 - Implemented an **IP-stride** and **complex-stride prefetcher**.
- **Distributed key-value store**
(CS 744: Design and Engineering of Computing Systems | **Prof. Purushottam Kulkarni**) (Aug'24-Dec'24)

- Designed and implemented a distributed key-value store with caching and evaluated the effectiveness of caching.
- **Brain tumor image segmentation using UNet++**
(CS 725: Foundations of Machine Learning | Prof. Preethi Jyothi) (Aug'24-Dec'24)
 - Used different augmentations, such as vertical and horizontal techniques, to improve brain tumor segmentation on the BRATS'2018 dataset.

INTERNSHIPS

- **Project Intern | Indian Institute of Technology, Kanpur** (Feb'24-June'24)
 - Worked on fine-tuning of LLMs for domain-specific applications.
 - Compared **QLORA, LORA, and RAG** methods for adapting **Llama-2** to cybersecurity tasks.
 - Built retrieval-augmented generation with vector databases to reduce model hallucinations.
 - Deployed Llama-2 model using **Pytorch, Hugging Face, and LangChain** for inference.
- **Project Intern | Indian Institute of Technology, Guwahati** (July'23-Jan'24)
 - Conducted comprehensive analysis of NEWS data related to military operations in China, creating a bespoke dataset with advanced NLP techniques.
 - Developed an efficient web application using **NextJs, FastAPI, Vercel, and Render** to provide intuitive insights for informed decision-making.

TECHNICAL SKILLS

- **Programming & Scripting Languages:** C, C++, CUDA, Python, Bash
- **Development & Debugging Tools:** GDB, Valgrind, Git, Vim
- **Profiling Tools :** NSight Compute, NSight Systems , Perf , Intel VTune
- **Simulation Tools :** GPGPUSim, VulkanSim
- **Frameworks & Libraries:** ReactJS, NextJS, Flask, Pytorch Numpy, Pandas, Scrapy

TEACHING

- **Head TA | CS230/231: Digital Logic Design and Computer Architecture** (Jul'25-Nov'25)
- **Head TA | CS101: Computer Programming and Utilization** (Jan'25-May'25)
- **Head TA | CS230/231: Digital Logic Design and Computer Architecture** (Jul'24-Nov'24)

EXTRACURRICULAR ACTIVITIES

- Volunteered as a TA for a computer architecture workshop organized by INAE.